

Session 06 Phase 3 – Master Results Tables

July 31, 2026

What each condition means

Every row below is a different way of turning the model’s internal activity while it writes an answer into a single feature vector. Think of that internal activity as a trail of numbers – one per layer, per generated token – and each condition as a different way of boiling that trail down.

core_max The baseline. Takes the single highest value each internal signal ever reaches while generating the answer – a snapshot of the loudest moment.

q_velocity Instead of the activity itself, tracks how much it *changes* from one layer to the next – speed instead of position. Picks up on whether the model’s internal representation is still being revised as information moves through the network, not just where it ends up.

q_static The same kind of peak snapshot as core_max, but softer: uses a high/low percentile instead of the literal maximum/minimum, so one unusual token can’t skew the whole signal.

core_concat Staples core_max’s peak snapshot and q_velocity’s change signal together into one longer feature vector – “where it ended up” plus “how it got there.”

joint_tensor Instead of computing two signals separately and combining them afterward, fits one shared compression across the combined raw data from the start, so the signals can influence each other during compression rather than only being stapled together at the end.

triple_concat Staples all three single signals – core_max, q_static, and q_velocity – together. The “use everything we have” condition.

Model	Condition	TruthfulQA	TriviaQA	NQ-Open	TyDiQA-GP
LLaMA-3.1-8B	HARP (published, supervised)	88.5	92.9	89.4	86.6
	core_max	84.4	86.4	85.7	83.1
	q_velocity	86.2	85.8	84.6	82.6
	q_static	84.3	86.7	86.1	84.0
	core_concat	86.0	86.8 [†]	85.3	83.2
	joint_tensor	84.5	86.4	86.3	83.3
	triple_concat	85.5	87.0	85.7	83.2
Qwen2.5-7B	HARP (published, supervised)	88.1	92.8	84.0	88.4
	core_max	87.1	<i>pending</i>	78.8	88.3[‡]
	q_velocity	87.7	<i>pending</i>	78.2	87.0
	q_static	87.1	<i>pending</i>	79.1	87.5
	core_concat	87.8	<i>pending</i>	79.4	87.8
	joint_tensor	87.2	<i>pending</i>	78.5	87.0
	triple_concat	87.8	<i>pending</i>	79.3	87.9

Table 1: **Literature-comparable results: pooled AUROC (%) under HARP’s evaluation protocol** (train on 75% of known questions; test on the rest plus all fully-hallucinated questions), 5-seed mean, Random Forest readout. All non-gray rows are *our* unsupervised feature variants scored under this protocol; the gray row is the published score of HARP’s own supervised detector, copied from their paper for reference (not recomputed by us). Bold = best of our conditions per column (HARP row excluded). [†]On TriviaQA, logistic regression outperforms the forest: core_max 87.3 ± 0.2 , q_static 87.3 ± 0.3 , core_concat 87.3 ± 0.2 – our best literature-comparable numbers, all three effectively tied. [‡]Qwen/TyDiQA-GP is the first cell where an unsupervised condition of ours reaches the published supervised score: core_max 88.3 ± 1.8 against HARP’s reported 88.4, a gap of 0.1 points, well inside our seed-to-seed spread.

Model	Condition	TruthfulQA	TriviaQA	NQ-Open	TyDiQA-GP
LLaMA-3.1-8B	core_max	81.77	81.97	76.10	84.48
	q_velocity	83.58	81.17	75.44	85.68
	q_static	83.28	81.97	76.72	85.26
	core_concat	84.35*	82.06	76.74	85.27
	joint_tensor	83.76	81.84	75.70	85.13
	triple_concat	83.28	82.27	76.01	85.76
Qwen2.5-7B	core_max	82.47	<i>pending</i>	74.45	87.39
	q_velocity	84.33	<i>pending</i>	74.88	88.50
	q_static	83.28	<i>pending</i>	75.32	86.88
	core_concat	84.43	<i>pending</i>	76.18	88.85
	joint_tensor	82.69	<i>pending</i>	75.87	87.11
	triple_concat	84.05	<i>pending</i>	77.61	89.14

Table 2: **Within-prompt AUROC (%), grouped 5-fold CV, Random Forest** – our stricter internal metric: among the 10 answers to the *same* question, can the detector rank the hallucinated ones above the truthful ones? No HARP reference row appears here because this metric is not reported in their paper. Bold = best per column, but note that a bolded lead is *nominal* unless marked *: on TriviaQA, NQ-Open, and TyDiQA-GP no condition’s within-prompt gain over core_max survives a paired bootstrap (all 95% CIs include zero; triple_concat’s TriviaQA lead, for instance, is $\Delta = +0.30$ pts, CI $[-0.14, +0.73]$). *core_concat vs. core_max on TruthfulQA remains, across all four completed datasets, the project’s *only* statistically significant within-prompt improvement (paired $\Delta = +2.57$ pts, 95% CI $[+0.03, +5.26]$, excludes zero). Separately, all four combination conditions *do* significantly beat q_velocity on TriviaQA on both metrics.

Model	Condition	TruthfulQA		TriviaQA		NQ-Open		TyDiQA-GP	
		RF	LR	RF	LR	RF	LR	RF	LR
LLaMA-3.1-8B	core_max	✓			✓	✓		✓	
	q_velocity	✓		✓		✓		✓	
	q_static	✓			✓	✓		✓	
	core_concat	✓			✓	✓		✓	
	joint_tensor	✓			✓	✓		✓	
	triple_concat	✓			✓	✓		✓	
Qwen2.5-7B	core_max	✓			<i>pend</i>	✓		✓	
	q_velocity	✓			<i>pend</i>	✓		✓	
	q_static	✓			<i>pend</i>	✓		✓	
	core_concat	✓			<i>pend</i>	✓		✓	
	joint_tensor	✓			<i>pend</i>	✓		✓	
	triple_concat				<i>pending</i>			✓	

Table 3: **Which classifier wins, under the HARP protocol** – ✓marks the higher of RF/LR (5-seed mean pooled AUROC) for that condition×dataset cell. RF sweeps TruthfulQA, NQ-Open, and TyDiQA-GP (18/18 LLaMA cells). TriviaQA flips: LR wins 5 of 6 conditions, RF holds only on q_velocity. See Table 1’s caption for the actual TriviaQA numbers (LR’s wins there are narrow, 0.2–0.9pt – not a landslide like RF’s wins elsewhere). Qwen2.5-7B’s first completed dataset reproduces LLaMA’s TyDiQA-GP pattern exactly: RF wins all 6 conditions, by 3.8–7.3 points.